



Supply Chain Climate Exposure

Greg Hall, Kate Liu, Lukasz Pomorski & Laura Serban

To cite this article: Greg Hall, Kate Liu, Lukasz Pomorski & Laura Serban (2023) Supply Chain Climate Exposure, Financial Analysts Journal, 79:1, 58-76, DOI: [10.1080/0015198X.2022.2129946](https://doi.org/10.1080/0015198X.2022.2129946)

To link to this article: <https://doi.org/10.1080/0015198X.2022.2129946>



© 2022 AQR Capital Management, LLC.
Published with license by Taylor & Francis
Group, LLC.



Published online: 08 Nov 2022.



Submit your article to this journal [↗](#)



Article views: 8644



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)

Supply Chain Climate Exposure

Greg Hall, Kate Liu, Lukasz Pomorski, and Laura Serban

Greg Hall is an executive director at AQR Capital Management, LLC, Greenwich, CT. Kate Liu is an executive director at AQR Capital Management, LLC, Greenwich, CT. Lukasz Pomorski is a managing director at AQR Capital Management, LLC, Greenwich, CT. Laura Serban is a managing director, AQR Capital Management, LLC, Greenwich, CT. Send correspondence to Lukasz Pomorski at lukasz.pomorski@aqr.com.

We propose an intuitive measure of supply chain climate risks, reflecting the fact that even a green company may have material climate exposure if its customers or suppliers face climate risks. Our measure captures price movements around climate news better than traditional climate data and shows performance patterns consistent with re-pricing of climate risks. The metric is more suitable for risk measurement than scope 3 emissions and requires raw data that is broadly accessible for large cross-sections of stocks and is of higher quality than currently available scope 3 data. We discuss applications for both portfolio and corporate decision-making.

Keywords: carbon emissions; climate; emissions offshoring; ESG; greenhouse gas emissions; scope 3; supply chain

Disclosure: No potential conflict of interest was reported by the author(s).

PL Credits: 2.0

Introduction

Climate-related risks and opportunities have become some of the most pressing priorities for investors. For most, climate-aware investing means managing a portfolio's carbon emissions, or its exposure to carbon reserves or green revenues. This is a natural starting point but is unlikely to offer a complete, holistic view of a portfolio's climate risk. Importantly, most traditional metrics will not capture exposures along the supply chain. The few climate metrics that have a supply chain component (e.g., scope 3 emissions) are usually motivated by carbon accounting rather than climate risk and are generally thought to be extremely noisily estimated.¹ We fill this gap by proposing a simple and intuitive measure of supply chain climate exposure. The measure reflects the strength of economic connections along the supply chain and is thus more likely to capture risks; it can also be readily computed by investors or their data providers for a broad cross-section of firms and using data that is verifiable and of arguably high quality.

To motivate our approach, we consider an example of a real-world payments services company from a sector (IT) and industry (IT services) that are rarely associated with climate risks.² In fact, based on standard climate data such as carbon intensity or carbon footprint, this company seems green even compared to its sector or industry peers. For example, its scope 1 and 2 emissions are lower than the industry and sector median, and its scope 3 emissions are in the

We thank two anonymous referees, Steffen Bixby, David Blitz, Nicole Boyson, Alfie Brixton, Jeff Dunn, Shaun Fitzgibbons, Andrea Frazzini, Luis Garcia-Feijoo, Jess Gaspar, Will Goetzmann, Antti Ilmanen, Lasse Pedersen, Roberto Rigobon, Luke Taylor, Dan Villalon, Eddie Watts, and Jing Zhang for helpful discussion and comments. AQR Capital Management is a global investment management firm which may or may not apply similar investment techniques or methods of analysis as described herein. The security mentioned herein was selected merely as an example and is not a recommendation to buy, sell, or hold. The views expressed here are those of the authors and not necessarily those of AQR.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way.

greenest quartile within its industry. Based on this information alone, we may conclude that the company has little climate risk exposure. However, our assessment may change once we look at its customers. For example, the company's 2012 10Ks state that its top three strategic relationships with major oil companies represented in the aggregate approximately 20% of its consolidated revenue. The large economic exposure to oil majors may indicate that the company might be indirectly, but possibly materially, exposed to disruptions and potential loss of revenue caused by climate-related risks.

This motivating example captures the economic intuition behind our framework. In essence, we propose measuring supply chain climate exposure as a business-weighted average of standalone climate exposure of a firm's customers and suppliers. For example, the downstream (customer-oriented) measure we analyze empirically in this paper is a revenue-weighted average of customers' own scope 1 + 2 carbon intensity. Our measure is higher for those firms that derive a large share of revenue from carbon-heavy customers, with the idea that climate risks harming those customers would also harm the firm's revenues. The lens of revenues makes it clear why supply chain climate exposure may be relevant for a firm's fundamentals (as would be any other exposure that potentially affects revenues: e.g., Kolay, Lemmon, and Tashjian [2016] document large costs incurred by suppliers to economically distressed firms).

While our empirical analyses focus on the downstream supply chain, the framework we present here can be employed to define an upstream metric: a cost-weighted average of suppliers' exposures. This measure would also capture an aspect of a firm's fundamentals, with the idea that firms that rely on suppliers exposed to climate risks may face disruptions to their supply chains when such risks materialize. Beyond the downstream and upstream measure, a third standalone metric that may round out the analysis is the portfolio company's own climate exposure, for example, its carbon intensity, a measure already very popular in investment practice (e.g., CFA Institute 2020, and the case studies included in that report).

We next empirically validate our downstream supply chain measure. This is a challenge rarely attempted in prior literature due to a paucity of data and the reality that the most severe climate risks may not have yet materialized. Nonetheless, we believe investors should insist on such analyses before they use a new metric in practice. We propose a series of such data tests, with the goal of not only increasing conviction in our supply

chain climate exposure measure but also giving investors a blueprint for testing other climate indicators.

We show that our measure has intuitive, but relatively low correlations with traditional climate data. More importantly, it helps explain stock price reactions to climate-related news sentiment, supporting our interpretation of climate risk exposure. The correlations we find (about 0.2) are similar in magnitude to the correlations of measures explicitly designed to track climate sentiment (e.g., Engle et al. 2020). In addition, our measure subsumes climate-related information that is contained in third-party measures (e.g., scope 1, 2, or 3 emissions). We also demonstrate that it exhibits intuitive patterns around those few climate-related events that have already materialized in our sample. For example, in the month following the G8 declaration to reduce emissions in 2009 (an example of positive climate news), the stocks we identify as green on our measure outperform the brown stocks by about 5%. Conversely, in the month following the US presidential election of 2016 (an example of negative climate news, given President Trump's vocal comments), green stocks lagged brown stocks by about 2%.

In addition, we found strong historical performance of long-short, market-neutral portfolios based on our measure. Historically, such portfolios realized a Sharpe ratio of about 0.85. This performance is not driven by broad country or sector bets (the country-adjusted within-industry version of the portfolio realized a Sharpe ratio of more than 0.7) and is not explained by standard academic factor models (the portfolio has annualized alphas of 6.5% to 9%). This is consistent with the market pricing in supply chain climate risks, possibly because they are deemed to be increasingly material or because investors have less appetite for holding these risks. At the same time, we show that downstream supply chain exposure also predicts fundamentals such as earnings or sales surprises, beyond what is captured by traditional quality, fundamental, or price momentum signals. This is consistent with our measure capturing a new facet of quality and potentially generating forward-looking returns, similar to other quality signals.

Finally, and importantly, the supply chain measures we propose are not meant to be a replacement for scope 3 emissions data. They are related and can answer similar questions but differ in their explicit focus on financial risk exposure. For example, a company may have zero scope 3 emissions but still score very poorly on our downstream measure because it

derives material revenue from customers with high climate exposure and vice versa. Another clear difference is that scope 3 isolates those supply chain emissions that are attributable to the company's products, making it a preferred measure for carbon accounting purposes. In contrast, our measure may double-count a company's emissions (e.g., a customer firm's overall emissions are counted separately for each of the firm's suppliers), which is not ideal for carbon accounting but in our view is acceptable as it leads to a more comprehensive climate exposure risk measure.

Given the strong interest in scope 3 emissions, Figure 1 compares scope 3 and our downstream supply chain measure and the key ingredients necessary to compute them. The arrows illustrate the focus of each metric: scope 3 captures emissions *from* suppliers *into* the product and *from* the product to customers; our measure instead depends on the economic revenue *from* customers to the company and *from* the company to suppliers.

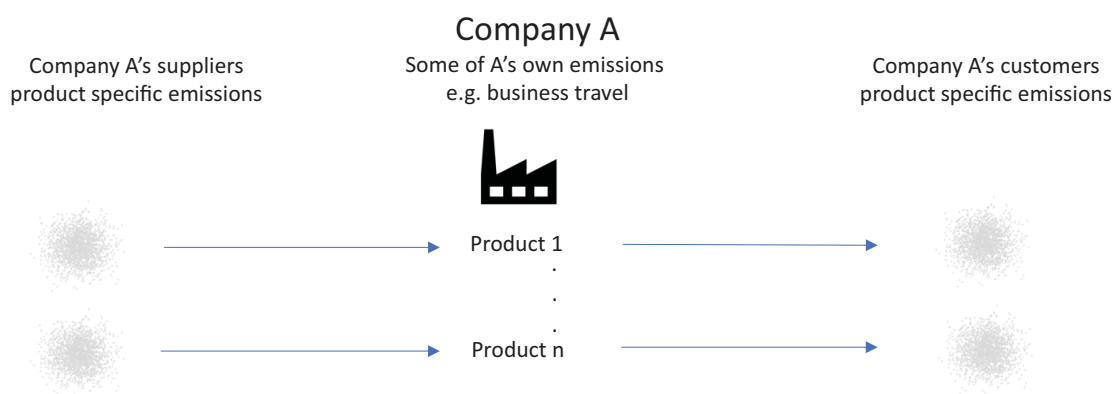
Practical Applications of the Supply Chain Climate Exposure Measures

The downstream supply chain measure we analyze in detail in this paper and its upstream supply chain analogue (cost-weighted average exposure of a firm's suppliers) lend themselves most naturally to portfolio context applications, but they are also useful for corporate decision-makers.

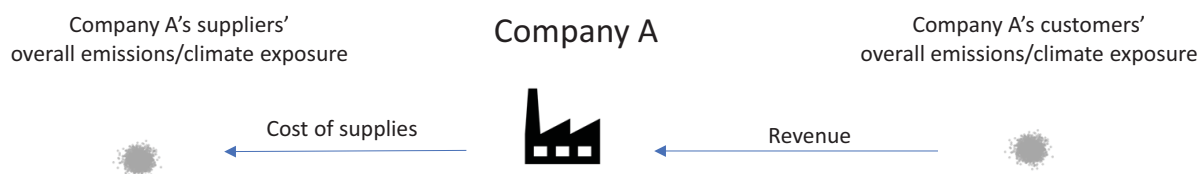
We motivate our framework by its potential to capture climate risks in the supply chain of a portfolio company and in the sections below validate this claim empirically. Hence, the most obvious application is risk monitoring and measurement, for example, highlighting situations when a portfolio drifts toward stocks with increased supply chain exposure and possibly triggering a further investigation of such stocks. The measures we propose can also be easily incorporated as a linear constraint in portfolio optimization, for example, targeting a

Figure 1. A Simplified Comparison between Scope 3 Emissions and Downstream Supply Chain Climate Exposure

Scope 3: accounting for the emissions traceable to a company's products and services



Supply chain climate exposure: accounting for the strength of economic links with, and overall emissions/climate exposure of a company's customers and suppliers



Note: The schematic shows which data inputs are relevant for each climate measure, highlighting the product-centric nature of scope 3 emissions and the economic linkage-centric nature of the downstream supply chain climate exposure measure proposed in this paper.
Source: AQR. For illustrative purposes only.

specific reduction in exposure versus a benchmark. Some investors may also decide to formally incorporate them into a risk model, with the goal of hedging out climate-type risks, possibly along the lines suggested in Engle et al. (2020). In addition, our finding of return patterns associated with the downstream measure suggests that it may potentially improve an investor's investment view, for example, by enhancing the understanding of a company's quality profile. We anticipate that such applications may be of interest to both discretionary and systematic investors.

Moreover, the data inputs underlying our approach are broadly available to investors and are of relatively high quality, resulting in excellent data coverage of our measure. These features support the use of the measure for reporting purposes, allowing comparisons across mandates or even asset classes (because the measure is at the issuer level, it can inform both equity and credit allocations).

Our indicator may also be useful in stewardship, for example, by identifying companies with abnormally high supply chain exposures as potential targets for engagement or by measuring the progress that portfolio companies make in engaging with their own supply chain partners. This may be particularly relevant for emissions offshoring, which occurs when a company switches from making the most carbon-intensive components inhouse to buying them from suppliers instead.³ When such components are produced using company-owned assets, they increase the company's scope 1 emissions; when they are bought from suppliers instead, they only affect scope 3 emissions. Because most investors today focus on scope 1 and 2, emissions offshoring may artificially increase investors' assessment of how green a portfolio company is. In principle, this behavior would be apparent in a firm's scope 3 emissions, but as we mention above, such data is unfortunately very noisy (e.g., Andersson, Bolton, and Samama 2016; Callan 2020; Cheema-Fox et al. 2021; Ducoulombier 2021). Our measure can at least partially fill this gap by helping identify companies that dramatically reduce their scope 1 emissions but at the same time look increasingly brown on our measure.

Finally, we believe our measure will be useful not only to investors but also to the portfolio companies. Corporations face pressure to report environmental, social, and governance (ESG)-type information, increasingly including information related to their supply chains, e.g., scope 3 emissions. Unfortunately, estimating product-related emissions is a daunting challenge. For example, in its 2020 Environment

Report, IBM (2020) stated that *"the assumptions that must be made to estimate Scope 3 emissions in most categories do not enable credible, factual numbers."* Busch, Johnson, and Pioch (2022) not only find large inconsistencies in scope 3 estimates but also find that such inconsistencies actually increase over time, suggesting that the growing investor and perhaps also regulatory interest in this measure has not yet led to any unified reporting. Ducoulombier (2021) points out that reporting on scope 3 remains largely voluntary today (even if some important regulators such as the U.S. Securities and Exchange Commission may require it in the near future), that existing reporting standards do not support comparisons between firms, and that data providers producing scope 3 estimates typically take insufficient consideration of firm-level circumstances. In contrast, our measure may not only be more material to a company's risks (as we argue above), but it will also be far easier for a corporation to compute and will be more comparable across companies. Corporations know precisely who their customers and suppliers are and how much revenue and cost they account for. They still need the data on the overall emissions of a customer or a supplier, but that data can be estimated more precisely: Indeed, unlike scope 3 emissions, they are self-reported by the majority of companies in broad indexes such as MSCI World (e.g., Bixby, Brixton, and Pomorski 2022; Bolton, Kacperczyk, and Samama 2022). In addition, an important advantage of our approach is that it can be adjusted to address at least some of the data weaknesses that are a perennial problem in ESG. For example, when customers' scope 1 + 2 carbon emissions data is not available (perhaps because they are private firms that do not report emissions), one could reasonably assume that their own climate exposure is in line with that of their peers. For example, even a simple regression of scope 1 + 2 emissions on industry dummies explains more than 40% of the variability in emissions. This flexibility extends the resulting data coverage for an investor and makes it easier for a corporate entity to compute its own supply chain climate exposure.

Data

For our empirical analyses below, we focus on large-cap developed stocks, roughly equivalent to the holdings of the MSCI World Index. Importantly, we do not restrict the universe of customers or suppliers of such stocks: For example, if a global developed firm in our sample has emerging markets customers, we

will utilize data on such customers to assess the firm's supply chain climate exposure.

To measure downstream supply chain climate exposure, we need two core pieces of data. First, we need data on the commercial linkages between a company and its customers. This data has been popularized by academic papers going back to Cohen and Frazzini (2008) and Menzly and Ozbas (2010). The supply chain data we utilize here is blended from multiple third-party vendors. These vendors use data from regulatory filings, earnings transcripts, company and industry publications, and news sources to comprehensively track global supply chains and the strengths of linkages. We merge vendor data to maximize coverage and precision. The linkages data we utilize provides the best coverage we are aware of for our universe of large-cap developed stocks: For example, as of December 2021, we can identify at least one customer for 87% of the stocks in our sample.

Ideally, the data would reflect not only a commercial link between two companies but also the strength of this link, for example, the fraction of a firm's revenue from a particular customer. For instance, Cohen and Frazzini (2008) collect this type of revenue data, leveraging disclosure regulations in the U.S. market (companies must report customers accounting for more than 10% of revenues). We have at least partial data on revenues for 58% of the companies with any linkage data. While this coverage falls short of 100%, we use state-of-the-art data, utilized in multiple other research papers, and are not aware of a dataset with better systematic coverage. Moreover, even with imperfect coverage, we capture the more material economic linkages (e.g., in the U.S., where these must be reported to investors). Conditional on having revenue figures for at least one customer, our data captures 19.4% of total revenues for the average firm.

The second piece of data we need is a measure of a customer's (or a supplier's) own climate exposure. In the analyses below, we opted for a simple measure that we believe will resonate with investors: scope 1 + 2 carbon intensity of customers and suppliers, as reported in the Trucost database. We use this as an intuitive indicator of which customers and suppliers are relatively greener or browner. We do not claim that this simple indicator is necessarily the "best" measure of climate risk exposure. We expect that investors interested in our approach will instead substitute their favored instruments, whether carbon emissions from another vendor such as MSCI or climate VaRs, Implied Temperature Rise scores, and so

on. Our methodology easily accommodates such measures.

Finally, we use a variety of standard returns and financials data from XpressFeed and a variety of ESG-related metrics from MSCI and Trucost, as indicated below. Overall, our sample spans the period of May 2009 through December 2021.

Missing Data and Limitations of the Supply Chain Climate Exposure Measure.

Missing data is an inevitable part of research, and how missing data is dealt with may meaningfully influence the analysis. We assume that a company with missing carbon intensity data has similar carbon intensity as its same-sector peers. This is preferable to ignoring customers or suppliers with missing data and effectively giving them a free pass even when they are in sectors that are very pollutive. As we discuss below, when dealing with missing data on the fraction of revenue companies have from a given customer, we impute the strength of the relationship using both the number of customers with missing revenue data and their market cap, so that same-sized customers account for the same fraction of revenue.

One potential weakness of our approach is that we may not observe a customer-supplier linkage in the first place (We note that if portfolio companies self-report on such measures, they of course have full information about their own economic links). To partially address this, we normalize the data we have to 100%. For example, if the data at our disposal captures only 70% of a firm's revenue, we rescale it so that it sums up to 100%. This implicitly assumes that the missing firms are as green as a typical customer on which we do have coverage. This is not ideal because, in some circumstances, the missing data may be statistically biased. For instance, our data does not include companies' retail customers. One might expect that retail customers have a different, perhaps "greener" profile than that of a typical corporate customer. Unfortunately, we do not know how much of the missing data covers retail versus corporate customers, making it challenging to adjust the measure. Similarly, companies that may be offshoring emissions, as we discuss in the Practical Applications section, may strategically seek suppliers less likely to be covered by data providers (e.g., private firms, or firms in emerging or frontier markets).⁴

In our assessment, data quality is sufficient for practical applications, in spite of the concerns above. Data on customer linkages may not be perfect, but it

features prominently in academic and practitioner research, so it seems fair to assume that much of the industry is familiar and perhaps even comfortable with it.⁵

Another potential weakness of our analyses below is that we use scope 1 + 2 emissions to gauge customers' own climate exposure. This is only one of the many climate metrics investors may consider, but our approach easily extends to other metrics. To the extent an investor is comfortable using a different variable to gauge climate exposure of a portfolio company, it stands to reason that the investor should be comfortable using the same variable to gauge supply chain climate exposure.

Constructing the Downstream Supply Chain Climate Exposure Measure

To make our downstream measure more tangible, we start with a hypothetical example, showing how our measure is computed. Consider a firm with four customers, whom we label with subscript j below (so, $j = A, B, C$, or D for the four customers). As mentioned above, we do not restrict the customers to be in the same investment universe: For example, the firm we study may be a large-cap developed stock, while its customers may be small-cap and emerging markets companies. To compute our measure, we need to know each customer's climate exposure, C_j . In our applications below, C_j is customer j 's scope 1 + 2 carbon emissions intensity. In our example, we assume we have this data for each of our hypothetical customers; in practice, missing data could be replaced by industry or sector averages. We also need data on the fraction of revenue each customer j represents. To make our example

realistic, we assume that we have that data for two customers, A and B , and that the data is missing for customers C and D . Table 1 summarizes the data for our example.

We first impute the missing revenue data for customers C and D by normalizing the total revenue to 100%. This implicitly imposes the assumption that the company only has four customers, that is, that we are not missing any customer linkage. For most firms in our sample, this is likely to be an approximation only, but we would expect that our data would include at least the most important commercial relationships of each firm. For example, for U.S. companies, we know all relationships that account for more than 10% of revenue, given the regulatory requirement to report them. In Table 1, we assume that customers C and D are equally important; in our empirical tests below, we impute the fraction of revenue using both the number of customers and their size, arriving at higher imputed revenue for customers with a higher market cap.

Once we have the data on both customers' own climate exposure, C_j , and their relative revenue, w_{ij} , our downstream supply chain climate exposure measure for company i is simply the revenue-weighted climate exposure, as per Equation (1):

$$\text{Downstream Supply Chain Climate Exposure}_i = \sum_j w_{ij} * C_j, \quad (1)$$

where in our empirical tests below C_j is customer j 's scope 1 + 2 carbon intensity, defined as scope 1 + 2 emissions divided by sales. We normalize by sales rather than using overall emissions in tons, as the latter would skew the metric toward large companies (a large company will generally have higher overall carbon emissions).

Table 1. Computing Downstream Supply Chain Climate Exposure for a Hypothetical Company, i , with Four Customers, $j = A, B, C$, and D

	Customer's Carbon Intensity (C_j)	% Revenue from Each Customer	Imputed % Revenue (w_{ij})
Customer A	200	40%	40%
Customer B	100	30%	30%
Customer C	100	NA	15%
Customer D	50	NA	15%

Source: AQR. For illustrative purposes only.

We focus on the downstream (customer) version of the supply chain measure in our subsequent discussion. The upstream (supplier) version can be computed analogously:

$$\text{Upstream Supply Chain Climate Exposure}_i = \sum_j v_{ij} * C_j, \quad (2)$$

where v_{ij} is the fraction of business that company i does with supplier j and C_j is supplier j 's scope 1 + 2 carbon intensity. The resulting metric is a cost-weighted climate exposure of a company's suppliers.

Summary Statistics of Downstream Supply Chain Climate Exposure

We begin by comparing our supply chain measure to the typical carbon emissions metrics: scope 1, 2, and 3 carbon intensity. Our analysis deals with large-cap developed equities, so not surprisingly the resulting data coverage is excellent. Our supply chain measure can be computed for 93% of the index weight in the average sample quarter and covers 96% of the index toward the end of our sample; standard intensity measures, as reported in the Trucost data, cover 87% of the index weight on average and 95% toward the end of our sample. While the resulting coverage happens to be similar for our measure and for standard carbon intensity data, we note that this is not guaranteed. We can compute our metric even for those stocks that are not covered by Trucost, as long as we can identify at least some of their customers. In fact, we can compute our measure even if we do not have such customers' own carbon intensity data, because we can leverage sector or industry membership of these companies and use sector or industry averages as proxies. For example, an IT services company that received 10% of its revenue from an Oil and Gas customer is likely exposed to supply chain climate risks, whether we explicitly have the customer's emissions data or not. These features of our approach may be particularly important in those investment universes where the coverage of standard emissions data is poor, for example, for small-cap equities or for private issuers in credit.

Figure 2 presents supply chain and standard emissions statistics for both a typical MSCI World constituent and for the MSCI World Index overall. As explained in the earlier sections, our downstream supply chain climate exposure measure is a revenue-level average of the carbon intensities of each firm's customers. The median stock's downstream exposure is smaller than, but relatively closer to, scope 3

emissions than to either scope 1 or scope 2.⁶ In contrast, at the index level, estimated supply chain climate exposure is meaningfully higher than that implied by traditional emissions data. This is driven by larger firms having higher supply chain exposure (indeed, we see a strong correlation between supply chain exposure and market capitalization).

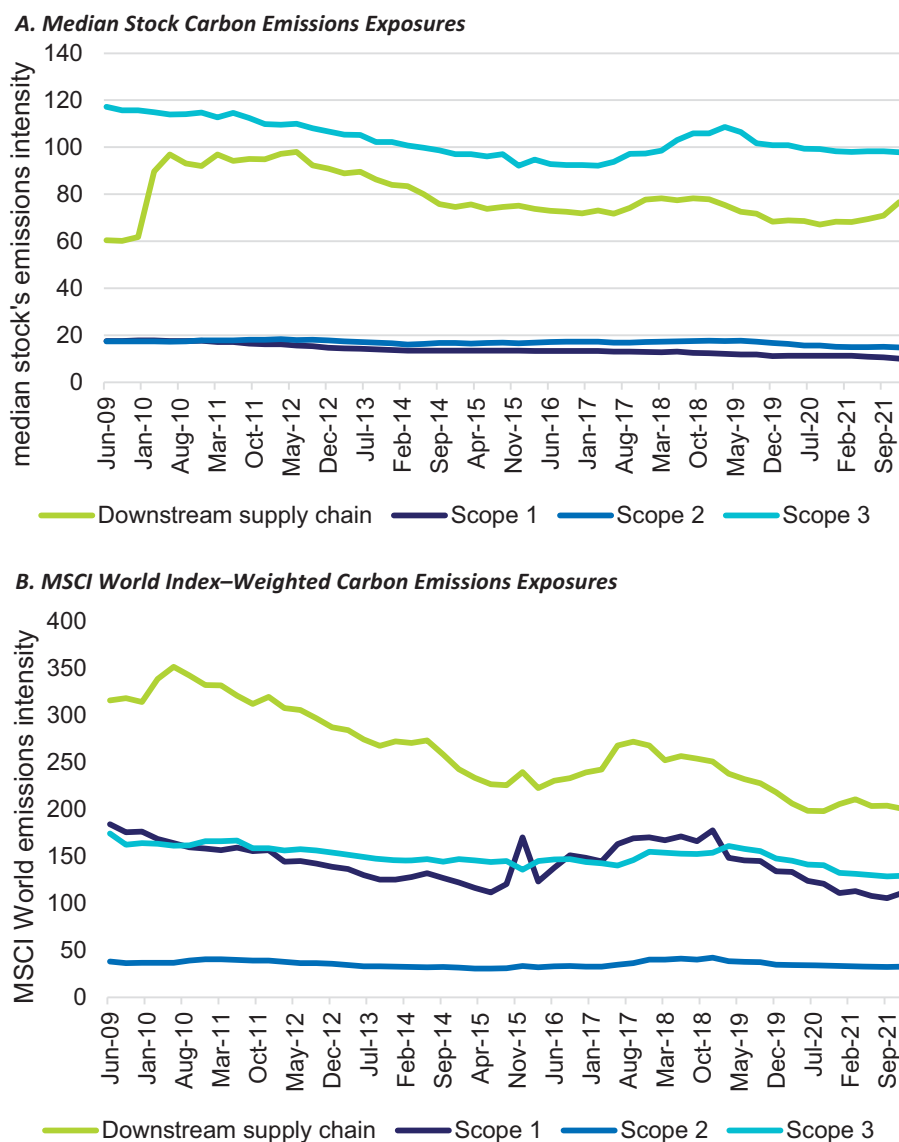
Figure 2 suggests that our measure is correlated with standard carbon emissions metrics. For example, we observe correlations of 0.1 to 0.15 between our measure and scope 3 emissions. This may seem low given that the two metrics both focus on the supply chain, but—as we point out in the introduction—they capture fundamentally different dimensions. More surprisingly, perhaps, our measure correlates with scope 1 emissions, meaning that firms that produce more emissions from company-controlled sources also have customers who emit a lot. In this case, the correlations increase to a more meaningful but still relatively low level of 0.3 to 0.4. This is partially because our measure depends on customers' scope 2 emissions: At least some of these will be the same as scope 1 emissions of a given firm (in line with this intuition, the correlation is about twice as high for Utilities than for firms in other sectors).

We now turn to the distribution of our measure across sectors and geographies, as presented in Figure 3. For ease of presentation, we show the results in terms of the contribution of each sector (in Panel A) and each region (in Panel B) to the overall index level. For comparison, we also show the index weight of each sector and region. We compute the contributions for each quarter in our sample and then show the contribution for the average quarter in Figure 3.

In Panel A, we see that our downstream measure spikes for Utilities, Energy, and Materials, the three sectors that are also associated with high scope 1 emissions, suggesting that their (revenue-weighted) average customer is also carbon-intense. This may not be surprising: Companies that are important customers for these sectors, and therefore use a lot of energy, oil, gas, or minerals, are likely companies that also have substantial greenhouse gas emissions.

The distribution by sector also shows that the downstream carbon metric may capture different information than scope 1, 2 and 3 carbon intensity. For example, we see a relatively higher downstream exposure for Industrials than one may expect from traditional emissions data. Similarly, Financials have relatively larger downstream supply chain climate

Figure 2. Carbon Emissions Statistics over Time

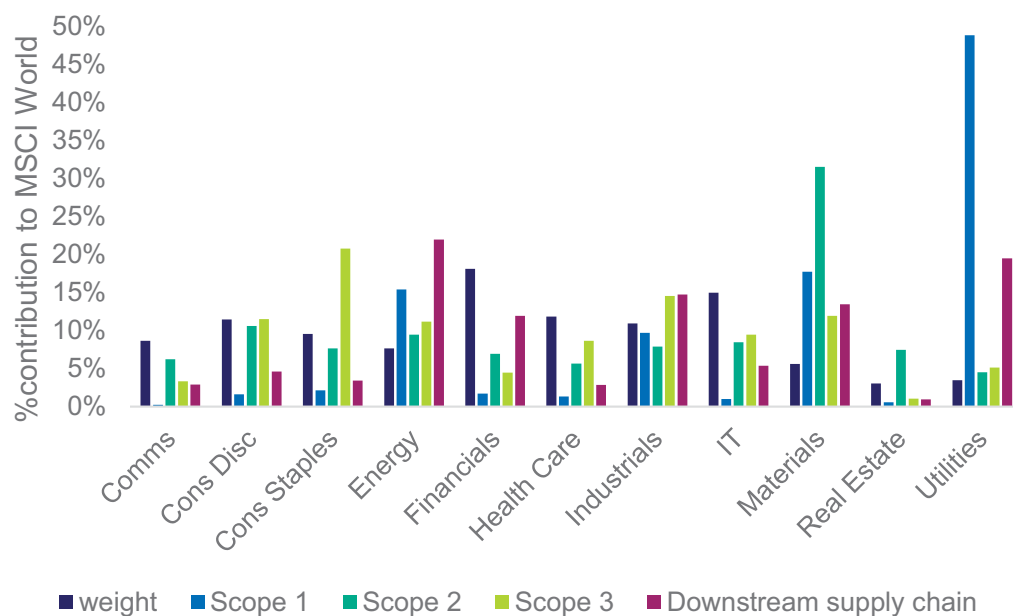
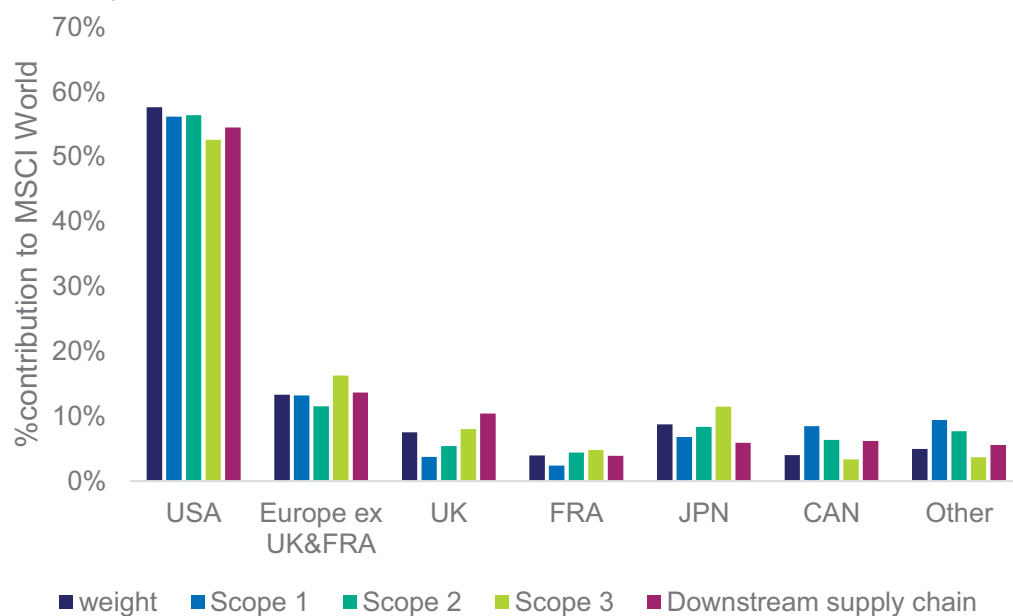


Note: The downstream supply chain climate exposure and scope 1, scope 2, and scope 3 emissions intensity for the median MSCI World constituent in Panel A and for the overall MSCI World Index in Panel B for the period of June 2009 to December 2021.

Source: AQR, MSCI, Trucost.

exposure relative to scope 1, 2, and 3 carbon intensities. This suggests that some industrials and financial companies provide goods and services to customers who may be subject to significant climate risk, despite these companies' stand-alone processes being relatively green. In contrast, downstream exposure is relatively less pronounced for Consumer Staples, at least compared to the scope 3 emissions of that sector. This is perhaps surprising. As we

mentioned in our data section, our customer data does not include data on retail customers who may be relatively more important for Consumer Staples; however, for this to explain the patterns in Figure 3, retail customers would need to be more carbon-heavy than corporate customers. Finally, we note that supply chain climate exposure is relatively less concentrated at the sector level than scope 1 exposure.

Figure 3. Sector- and Country-Level Contributions to MSCI World Index-Level Carbon Emissions Metrics**A. Sector Contribution to Various Carbon Emissions Metrics****B. Country-Level Contribution to Various Carbon Emissions Metrics**

Note: Each sample quarter, percentage contributions to MSCI World Index are calculated from each sector (Panel A) and each country or broader region (Panel B) as of December 31, 2021. The charts show the time series averages of these contributions. In Panel A, the historically deprecated Telecommunication is omitted from the chart.

Source: AQR, Trucost, MSCI.

Panel B of Figure 3 repeats the analysis by geography, looking at the countries with the highest weight in the MSCI World Index. We observe some variability related to size (e.g., USA stands out in absolute terms, but not when compared to the index weight of that region), but no significant concentration beyond that. A couple of countries stand out in that their downstream supply chain exposure is noticeably different than the traditional scope 1, 2, or 3 exposure. For example, the UK contributes 4% to 8% to MSCI World overall scope 1, 2, 3 weighted average carbon intensity but as much as 10% to the index's downstream supply chain exposure (for comparison, the average index weight of the UK is about 8%). On the flipside, Japan contributes less to index-level downstream exposure than it does to scope 1, 2, or 3 emissions, suggesting that it does business with relatively greener customers than its own emissions profile would suggest. Last, we see that commodity-based economies such as Canada contribute more to the downstream supply chain exposure than their weight in the index or their scope 3 emissions. It seems intuitive that commodity producers have carbon-heavy customers, but it is perhaps more surprising that they score as "brownier" on our supply chain measure than they do on scope 3 emissions. Canada's contribution to index-level downstream exposure is twice as high as its scope 3 contribution (Australia, while lumped into "Other" in Panel B, also contributes twice as much to the supply chain exposure as it does to scope 3). One explanation for this pattern is that the customers of commodity-producing countries are relatively browner not just because of those countries' own scope 3 emissions but also because of other sources of emissions; another is that scope 3 data may be missing some components of customers' scope 1 + 2 emissions that we explicitly account for here.

Validating the Measure

Our approach to measuring supply chain climate exposure is based on a clear and intuitive economic idea. However, before we can advocate practical use of any measure, we want to go beyond the intuition and to the extent possible to also verify it with data. Unfortunately, this is a significant challenge for ESG measures broadly speaking and for climate measures in particular. This is because the outcome variables we care about here—instances of climate risks affecting company valuations—are rare in historical data; the most severe climate events, whether related to physical or transition climate risks, may only occur in the coming years and decades. This may be true, but

it is a very poor excuse not to look at data. Markets do react to potential future risks, so climate risks may well affect historical stock prices even before they materialize. Realistically, climate may account only for a small fraction of a stock's total or even residual volatility—clearly less than, say, market beta, industry membership, and so on. For this reason, we sharpen the empirical tests below by conditioning them on climate-related variables, for example, looking at returns around climate-related news, when climate risk is more likely to be priced in or re-priced.

We first relate our downstream supply chain measure to more traditional climate data, including Environmental (E-pillar) scores and carbon emissions. A positive correlation with related third-party climate data would lend credibility to our measure. If we instead saw zero correlation with "standard" climate data, it would be difficult for us, or for the investor community, to accept this measure, no matter how plausible the intuition behind it. Second, we test whether downstream climate exposure correlates with climate sentiment. That is, we check whether the companies identified by our measure as relatively green or relatively brown experience returns, with intuitive signs, versus measures of market-wide sentiment following climate news. Third, we use an event study methodology to gauge the price behavior of stocks around meaningful macro-level climate news, as a function of stocks' downstream supply chain exposure. These tests may be circumstantial, but we believe that the preponderance of evidence justifies the use of our proposed metric in investment practice.

Comparison to Third-Party Climate-Related Metrics.

To compare our downstream supply chain climate risk measure with other climate data, we regress it on scope 1, 2, and 3 emissions intensity, as reported by Trucost, and on E-pillar and climate change theme scores, as reported by MSCI.

Table 2 shows that downstream supply chain climate exposure is strongly related to third-party measures of emissions, in most cases with the intuitive signs. For example, we expect a higher (riskier) downstream exposure for those companies that also have high carbon emissions intensity, at least in terms of scope 1 and 3 emissions.⁷ Similarly, we find intuitive signs for climate-related measures from MSCI ESG data: Higher downstream supply chain carbon exposure correlates with lower (i.e., less attractive) overall MSCI E score in regression (2) and lower Climate Change theme scores, the component of E that is most related to our measure, in (3).⁸ We pool the

Table 2. Explaining Downstream Supply Chain Carbon Exposure with Traditional Climate Data

	(1)	(2)	(3)	(4)
In CO2 intensity, scope 1	0.29** (56.08)			0.08** (14.43)
In CO2 intensity, scope 2	−0.09** (−8.63)			0.01 (1.76)
In CO2 intensity, scope 3	0.15** (13.03)			0.09** (6.83)
MSCI E-pillar score		−0.05** (−4.16)		0.00 (0.90)
MSCI climate change score			−0.04** (−5.40)	0.02** (3.93)
Sector + region indicator	NO	NO	NO	YES
Observations	77,194	84,249	65,727	55,970
R ²	0.280	0.011	0.010	0.461
Number of x-sections	51	51	36	36

Note: Estimates are from Fama-MacBeth regressions of the downstream supply chain carbon exposure (in logs) on scope 1, 2, 3 carbon intensity (in logs), MSCI E-pillar, and MSCI Climate Change theme. Regressions (1)–(3) do not include any additional explanatory variables; regression (4) also includes sector and regional indicator variables. The regressions are run quarterly from June 2009 to December 2021 for the developed large-cap universe (MSCI World); the table presents time series estimates *t*-statistics adjusted for heteroskedasticity and auto-correlation with Newey-West procedure with 4 lags.

*Indicates significance on a 5% basis and **Indicates significance on a 1% basis.

Source: AQR, Trucost, MSCI.

explanatory variables and add sector and region indicators in regression (4) and again find strong positive correlations with scope 1 and 3 emissions. This time, surprisingly, we find a positive rather than a negative coefficient on the MSCI Climate Change theme, counterintuitively suggesting that issuers with a higher (i.e., more attractive) climate change score also have higher downstream supply chain carbon exposure. The sign flip is likely driven by the correlations between the various climate variables; the interpretation of the coefficient on MSCI Climate Change score is more difficult when these correlated variables are included in the regression.⁹

Importantly, while we find strong and mostly intuitive correlations with typical ESG data, these third-party variables only capture a small amount of the variability in our measure. R^2 values in Table 2 range from 1% for high-level E and Climate Change scores to 20% to 30% when controlling for emissions-based measures. Even in our “kitchen sink” regression (4), where we also control for sector and region membership, traditional climate data and sector/country fixed effects capture less than half of the variability of our measure.

This combination of findings is encouraging. If we found no correlations whatsoever with third-party

climate data, this might mean that our measure simply does not capture any climate-related information. The third-party measures we use here, and similar measures we did not have access to for this paper, come from well-known, reputable data providers who work with some of the most sophisticated investors and consultants. These measures may be noisily estimated, but we posit that they capture at least some climate-related information, meaning that we should expect at least some correlation with our own metric.

At the same time, it is comforting that the correlation is imperfect. Had we found R^2 values close to 1.0, we would have concluded that our measure effectively conveys the same information as what is already captured by traditional providers. This would reduce the potential usefulness of our idea to those allocators who already subscribe to the standard climate data. Of course, finding correlations lower than 1.0 is not a sufficient condition for success: The low correlations might after all be attributable to white noise that carries no incremental information about climate. To test whether this may be the case, and to isolate the incremental information in our measure, we resort to sentiment analysis in the next section.

Correlation with Climate News

Sentiment. While the most meaningful climate risk events may materialize only in the future, their likelihood of occurrence and potential impact are certainly discussed in the news media today. We can leverage such discussions to validate our measure: If it indeed captures climate risk, then stocks with relatively higher exposure should do poorly when negative climate news arrives; stocks with relatively lower exposure should do well. The advantage of this approach is that we rely on the overall market, rather than on individual analysts, to assess which stocks are more exposed to climate risks. Of course, we do not observe such market assessment directly, but we can infer it implicitly from price movements around climate news events, as explained in Engle et al. (2020) and AQR (2021). Moreover, we can use this insight to check whether our measure captures information not yet contained in standard climate data. We do this by sorting stocks on a version of supply chain exposure that is orthogonalized to other climate data.

To carry out this analysis, we need data on climate sentiment. We utilize the Media Climate Change Concern index (MCCC) from Ardia et al. (2021), made publicly available from these authors, and measure the arrival of (signed) climate news by looking at the monthly changes to the index. This climate news sentiment data was used in other papers as well, for example, in Pastor, Stambaugh, and Taylor (2021), and was available until June 2018.¹⁰

To measure the relationship between downstream supply chain climate exposure and climate news sentiment, we form hypothetical factor portfolios that long 50% of stocks with low downstream supply chain climate exposure (we refer to these as “green stocks”) and short the other 50% of stocks (“brown stocks”). To avoid unintended bets, we form these portfolios in a country-neutral manner, with or without industry adjustments, and target zero beta (equity market neutral) and 7% ex ante volatility. As before, we use large-cap developed market stocks for this analysis. Figure 4 presents the (monthly) correlations between these factors and climate news sentiment.

Figure 4 shows that factors based on downstream supply chain climate exposure co-move with the climate news sentiment, with the positive correlation of 0.15 to 0.2. That is, the stocks we identify as greener outperform the stocks we identify as brown at times of increased climate concerns. Given the length of our sample (109 months), the *t*-statistic on this

correlation is 1.76. To put these correlations in context, we note that our supply chain measure does not utilize any news data. For comparison, portfolios explicitly optimized to correlate with climate news sentiment in Engle et al. (2020) also show out-of-sample correlations of about 0.2.

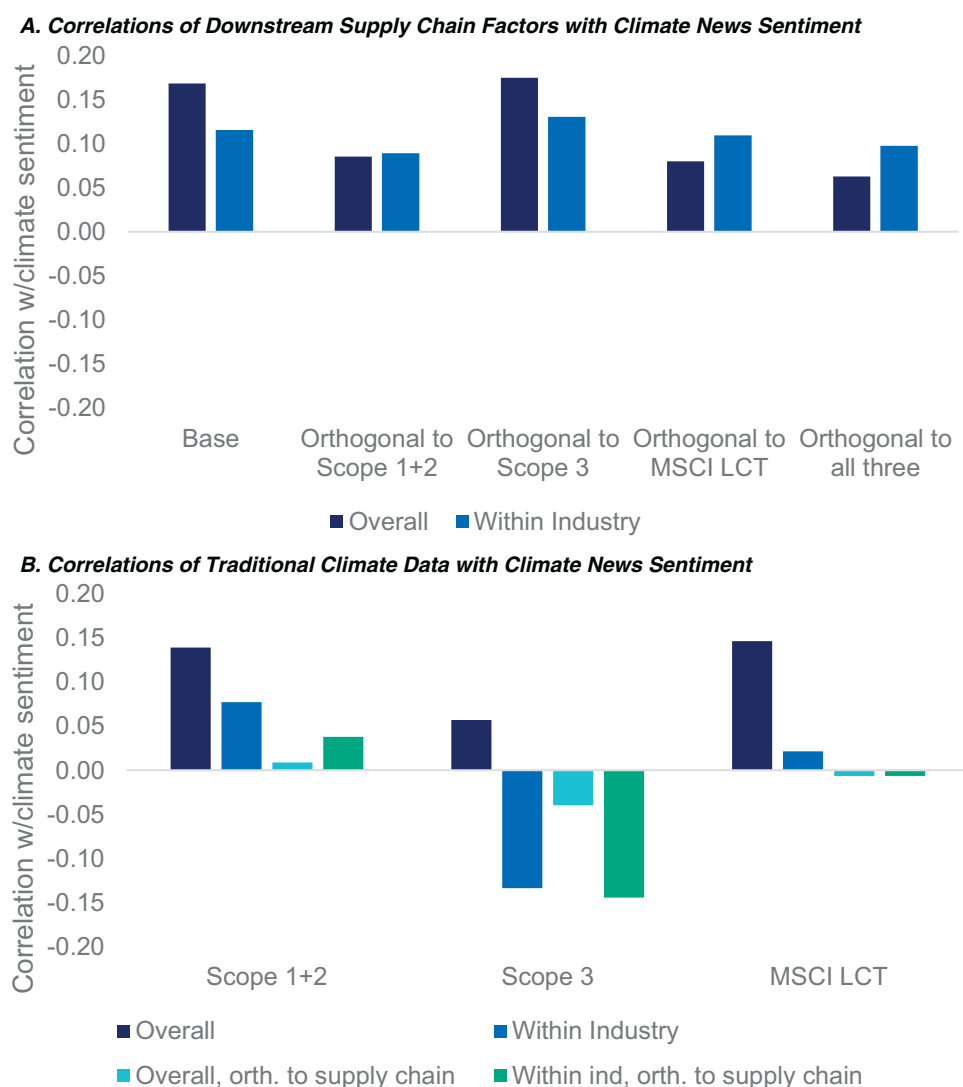
We believe that these positive correlations are an important validation of our measure, indicating that downstream supply chain climate exposure captures information that the market subsequently deems important in the climate context.

We still need to check whether such information might simply be a function of some relatively obvious characteristics, such as sector or industry membership (for example, as we saw above, Utilities stand out on our measure). To address this, Figure 4 also shows portfolios that are defined entirely within industry. We still see a positive correlation here, even if it is somewhat lower than before. Another potential concern is that our measure is to some extent correlated with traditional climate data (as we saw in the prior section) and the return correlations with climate news might be indirectly attributable to such third-party data and not to our downstream supply measure per se. Figure 4, Panel A shows that this is not the case: We find similar levels of correlations also for factors that were orthogonalized to scope 1, 2, and 3 emission and to MSCI’s Low Carbon Transition (LCT) scores.¹¹

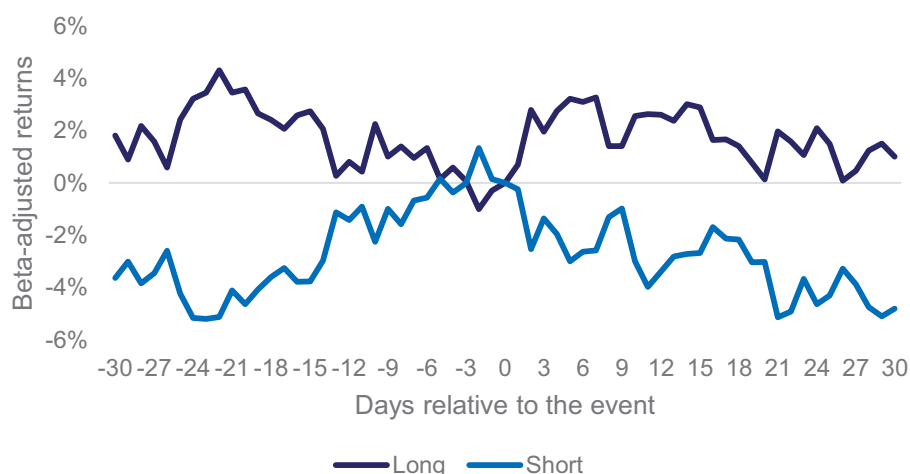
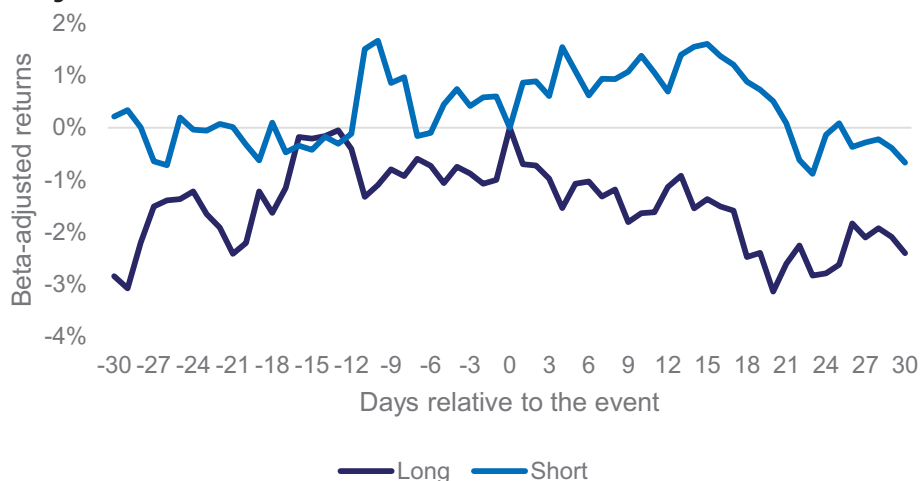
Strikingly, the reverse does not hold when we instead analyze factors based on traditional climate data in Figure 4, Panel B. On their own, factors based on greenhouse gas emissions, or on MSCI Low Carbon Transition scores, show correlations that are relatively similar to those in Panel A. However, as soon as we control for industry membership, these correlations disappear for LCT scores or even turn negative for scope 3 emissions. We still find low positive correlations for scope 1 + 2 emissions, but they decrease to almost zero when we orthogonalize the factors to our downstream supply chain measure. We conclude that the correlations between traditional climate data and climate news either are due to industry membership or are subsumed by our downstream supply chain metric.

Reaction to Macro-Level Climate Change News

Our last validation test relies on a simple event study methodology: We utilize the long-short factors based on downstream supply chain carbon exposure data, introduced above, and assess how they behave around major climate events. This

Figure 4. Correlation of Market-Wide Climate Sentiment with Factors Built Using Downstream Supply Chain Carbon Exposure

Note: The top panel shows monthly return correlations with the monthly changes in climate news sentiment index of Ardia et al. (2021) for factors based on downstream supply chain climate exposure overall and within-industry exposure as well as after removing the commonality with other typical climate-related data. The bottom panel shows similar correlations for a hypothetical long-short portfolio based on typical climate-related data, removing the commonality with the downstream supply chain measure, as indicated in the chart. The hypothetical portfolio ranks stocks on (negated) downstream supply chain carbon exposure; neutralizes country and, as indicated, industry exposures; removes market exposures based on ex-ante beta; and targets 7% ex ante volatility. This hypothetical portfolio is rebalanced monthly. Returns are gross of transaction costs, financing costs, and fees. Given the available climate sentiment data, the sample period is June 2009 through June 2018. Hypothetical performance results have many inherent limitations, some but not all of which are described herein. No representation is being made that any fund or account will or is likely to achieve profits or losses similar to those shown herein. Hypothetical performance results are presented for illustrative purposes only. Hypothetical performance is gross of advisory fees and transaction costs and includes the reinvestment of dividends. If the expenses were reflected, the performance shown would be lower.

Figure 5. Stock Reaction to Climate News Events**A. Example Positive Climate News Event: G8 Agrees to Reduce GHG Emissions by 80% by 2050****B. Example Negative Climate News Event: President Trump Withdraws the U.S. from Paris Agreement**

Note: Daily cumulative beta-adjusted returns to the longs and shorts from the downstream supply chain climate exposure factor, in the 30-day window surrounding two climate events: G8 committing to reducing emissions by 80% by 2050 on 9th July 2009 (Panel A) and President Trump withdrawing the U.S. from the Paris Agreement on 1st June 2017 (Panel B). The long and short side are beta-adjusted using the MSCI World Index benchmark. Universe of large-cap developed markets stocks, similar to MSCI World Index constituents. The long-short factor portfolio ranks stocks on their (negated) supply chain carbon exposure, neutralizes country and industry exposures, removes market exposures based on ex ante beta, and targets 7% ex ante volatility. This hypothetical portfolio is rebalanced monthly. Returns are gross of transaction costs, financing costs, and fees.

Source: AQR.

experiment is obviously related to the climate sentiment analysis above and may be noisier in that the information about such macro events may have been telegraphed to the media, and thus also the market, before the event itself took place. At the same time, this approach has the benefit of a straightforward intuition and is arguably more transparent than the more sophisticated natural language processing techniques behind sentiment measures. For this reason,

we illustrate this approach with two sample events, choosing one that is positive for climate (G8 committing on 9th July 2009 to reduce greenhouse gas emissions by 80% by 2050) and one that is negative (President Trump pulling the U.S. from the Paris Agreement on 1st June 2017). Given the construction of our factors, we would predict them to do well around positive climate news (i.e., we would expect the long leg of the factor to appreciate relative to

the short leg of the factor) and do poorly around negative climate news (longs depreciate relative to shorts). Figure 5 tests whether this is the case, showing market beta-adjusted cumulative performance of the longs and shorts in the 30-day window around the two events in question. The “0” date on the x-axis corresponds to the event itself.

The results in Figure 5 are more mixed, but we still observe some encouraging patterns. For positive news in Panel A, the short side of the factor (brown stocks) has negative market-adjusted performance after the event happens. The long side (green stocks) is more muted and noisier. For negative news, it is the long side that reacts more strongly, with more pronounced negative market-adjusted performance.

Of course, the two events we highlighted here are only a fraction of possible climate events one may consider. As we explained above, we showcase them here to illustrate the overall approach but would argue that the news sentiment analysis in the previous section is a more informative way to validate the measure.

Performance

Our final section discusses the performance of stocks sorted on the downstream supply chain climate exposure measure, utilizing the long-short portfolio we introduced in our climate sentiment analysis above. Hypothetical performance, gross of fees and transaction costs, is presented in Table 3.

We find strong performance for portfolios long in green stocks and short in brown stocks, identified using our measure. The portfolio formed ignoring stocks' industry membership realized a Sharpe ratio of 0.85 over this period. This strong performance is not subsumed by sector allocation (e.g., short Energy, long IT) as we also find a relatively similar Sharpe ratio for the industry-neutral version of the portfolio. As we show in Table 4, the performance is not explained by exposures to well-known investment factors such as Value or Momentum.

When the baseline portfolio is regressed on standard academic factors (Fama-French three- or five-factor models, potentially augmented with momentum and betting against beta [BAB]), it still shows economically and statistically significant abnormal returns of 6.5% to 9.4% p.a. As in Table 3, the within-industry results are slightly weaker but still strong in regressions (4)–(6). In regressions (7)–(9), we apply an even more conservative test, redefining factors to match the portfolio construction (within-country, within-

Table 3. Portfolio Performance for Longshort Portfolios Based on Downstream Supply Chain Climate Exposure

	Unadjusted	Within-Industry
Average return	7.4%	6.5%
Standard deviation of returns	8.7%	9.0%
Sharpe ratio	0.85	0.72
Beta to MSCI World	−0.07	−0.05
Hit ratio	56%	56%

Note: Universe of large-cap developed markets stocks, similar to MSCI World constituents. The long-short portfolio ranks stocks on (negated) downstream supply chain carbon exposure, neutralizes country and, when indicated, industry exposures, and removes ex-ante market exposures and targets 7% ex ante volatility. This hypothetical portfolio is rebalanced monthly. Returns are gross of transaction costs, financing costs, and fees. The sample period is June 2009 to December 2021. *Source:* AQR.

industry) and using the same universe of stocks as that used for the portfolio. We believe that such factors lead to a more conservative assessment in that they match the most important design choices of the portfolio being tested. The estimated alphas slightly decrease to 3.8% to 6% p.a. but remain economically sizeable and, with the exception of regression (9), significant at the 5% level.

Table 4 also indicates that the downstream supply chain factor tends to have a positive growth exposure, similarly to the green portfolio in Pastor, Stambaugh, and Taylor (2021); estimated coefficients on other investment factors are not consistent across specifications. We have verified that the value spread of the portfolio does not meaningfully widen over this period, as might be expected if our results were driven primarily by ESG investment flows. We also see at least as strong performance in the first half of our 2009 to 2021 sample as we do in the second half. In additional robustness checks, we find that the performance is not driven by a single country or region.

While this strong performance is arguably good news, we do not view it as strictly necessary for the validation of our measure. Our key research objective is to study (downstream) supply chain climate exposure, and it is at least theoretically possible that such exposures may not have strong performance implications. Having said that, we note that this performance can be consistent with the market pricing in

Table 4. Factor-Adjusted Performance of Downstream Supply Chain Climate Exposure (DSChCE) Long-Short Portfolios

	Unadjusted DSChCE Portfolio			Within-Industry DSChCE Portfolio			Within-Industry DSChCE Portfolio		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Alpha	6.48%** (2.70)	9.41%** (3.92)	8.92%** (3.41)	5.31%* (2.15)	4.95% (1.91)	6.55%* (2.35)	5.98%* (2.45)	4.85%* (1.97)	3.80% (1.46)
Mkt-rf	-0.02 (-0.52)	-0.05 (-1.12)	-0.05 (-1.03)	-0.01 (-0.22)	0.01 (0.16)	0.02 (0.31)	-0.02 (-0.35)	-0.03 (-0.52)	-0.03 (-0.59)
Size	-0.05 (-0.38)	-0.23 (-1.71)	-0.25* (-1.75)	0.17 (1.26)	0.19 (1.32)	0.26 (1.75)	-0.19* (-2.24)	-0.21* (-2.52)	-0.21* (-2.40)
Value	-0.39** (-4.58)	-0.56** (-4.34)	-0.53** (-3.56)	-0.44** (-4.99)	-0.57** (-4.08)	-0.51** (-3.17)	-0.40** (-5.13)	-0.27** (-3.03)	-0.18 (-1.24)
Profitability		-0.73** (-4.10)	-0.75** (-4.08)	-0.01 (-0.04)	-0.01 (-0.04)	0.05 (0.26)		0.18* (2.24)	0.17* (2.19)
Conservatism		-0.02 (-0.10)	-0.05 (-0.23)	0.31 (1.34)	0.31 (1.34)	0.33 (1.37)		-0.14* (-2.06)	-0.15* (-2.15)
Momentum			0.02 (0.24)			0.13 (1.27)			0.07 (0.44)
BAB			0.04 (0.36)			-0.23 (-1.88)			0.11 (1.19)
Observations	151	151	151	151	151	151	151	151	151
R ²	0.142	0.233	0.235	0.151	0.162	0.184	0.160	0.208	0.217

Note: Monthly returns of the downstream supply chain climate exposure long-short portfolio are regressed on the factors indicated in row labels. Regressions (1)–(3) use the portfolio defined both across- and within-industry; in (4)–(9), the portfolio is based on within-industry supply chain exposures only. Regressions (1)–(6) use academic factors (Market, Size, Value, Profitability, Conservatism, and Momentum are Mkt-rf, SMB, HML, RMW, CMA, and UMD, respectively from Ken French's Data Library; BAB is from the AQR Data Library), and regressions (7)–(9) use the within-industry versions of those factors, defined using equivalent stock characteristics (e.g., market cap for Size; book-to-price for Value, etc.). Alphas are annualized, and t-statistics are shown in parentheses. The sample period is June 2009 to December 2021.

* Indicates significance on a 5% basis and ** indicates significance on a 1% basis.

Source: AQR, Ken French's Data Library.

Table 5. Regressions Relating Quarterly EPS and Sales Surprises to Downstream Supply Chain Climate Exposure

Additional Controls	Unadjusted		Industry Demeaned	
	EPS	Sales	EPS	Sales
No additional controls	9.30**	5.64**	3.83**	1.72
Analyst revisions	7.77**	5.02**	3.04**	0.79
Analyst revisions + Profitability + Δ Profitability + Momentum	9.24**	5.55**	3.38**	1.03

Note: Each quarter, EPS and Sales surprises, as indicated in column headers, are regressed on a given firm's downstream supply chain climate exposure and additional controls prior to the earnings announcement, as indicated by row labels. The table shows the t-statistics of the coefficient on supply chain climate exposure, obtained from the time series of cross-sectional regressions using the Fama-MacBeth method. The t-statistics are adjusted for heteroskedasticity and auto-correlation with Newey-West procedure with four lags. Profitability is defined as return on net operating assets; Fundamental Momentum is defined as 1-year change in Profitability; and Price Momentum is based on the total return of stocks in the past 12 months excluding the most recent month. The data in the right panel is industry-demeaned. The sample period is June 2009 to December 2021. Universe is large-cap developed markets stocks, similar to MSCI World constituents.

*Indicates significance on a 5% basis and ** Indicates significance on a 1% basis.

Source: AQR.

climate risks. Arguably, markets have paid increasingly more attention to climate, so as long as they care about the downstream supply chain exposure, the prices of issuers with high exposure may have fallen relative to the prices of issuers with low exposure, leading to the positive returns in [Tables 3 and 4](#). Pedersen, Fitzgibbons, and Pomorski (2021) show how such repricing occurs in response to changes in the preferences of, or the information available to, market participants.

At the same time, there are alternative explanations: For example, our climate exposure measure might be indicative of higher production efficiency and higher overall quality, as suggested in a related context in Garvey, Mohanaraman, and Nash (2018). While traditional measures of quality (e.g., profitability, or even emissions, as in Garvey, Mohanaraman, and Nash 2018) relate to a specific portfolio company, it is possible that quality extends also across the supply chain. For example, firms that have customers of higher quality, including in the dimension of carbon emissions, may be well positioned going forward because their customers are less likely to experience adverse risk events. To test such alternative explanations of abnormal performance, we turn to [Table 5](#), where we assess whether downstream supply chain climate exposure can help predict companies' quarterly EPS and sales surprises. To do so, we run Fama-MacBeth regressions of surprises at quarterly earnings announcements on a given company's downstream supply chain exposure and a range of controls, meant to capture analyst revisions, profitability characteristics, fundamental momentum, and

price momentum ahead of the earnings announcement.

[Table 5](#) indicates that companies with “greener” downstream exposure on average experience larger EPS surprises than other companies. This effect persists also when controlling for analyst revisions and price and fundamental momentum, with or without industry adjustment. We see similar results for sales surprises, although we lose significance when variables are industry-demeaned. While this relationship with fundamentals might be explained by risk exposures, we believe it is more likely to reflect a facet of Quality, perhaps related to the resource efficiency of a company along its supply chain. If so, then the strong historical returns are consistent with the premium we observe for other Quality-type variables, say, profitability. In this case, we may expect such premia to persist going forward as well.

Conclusions

We propose a novel method to measure a firm's climate risk exposure in the supply chain, leveraging the firm's economic linkages. We make a simple but, to our knowledge, not-yet-proposed point that such risks can be modeled as revenue-weighted climate exposures of a company's customers or a cost-weighted climate exposure of its suppliers. Our framework can be utilized with one's preferred measure of exposure, potentially capturing physical climate risks, transition opportunities, green revenues, and so on. In our empirical validations, we find that

downstream supply chain climate exposure captures co-movement of stock returns around climate news events, largely subsuming relevant information from other traditional climate data, and that its performance is consistent with the market (re)pricing climate risk exposures.

Our measure is similar to, but not a replacement for, the often-discussed scope 3 emissions. It captures somewhat different information that we argue may be more material for issuers' financial risks. It also has a meaningful advantage of being easier to assess, using data that is already broadly familiar to the investment community. While our measure is not available "off the shelf" from third-party vendors, the straightforward methodology we explain in detail in this article can be implemented in-house by at least

some investors. Moreover, while we focus on public equities in this article, investors can leverage our idea also in other asset classes. This is most obvious in credit, but also holds in sovereign fixed income, where one could consider trade-weighted exposure to countries that produce more emissions or are otherwise exposed to climate risks.

The flexibility of our intuitive methodology, good data coverage, and the additional validations we present here bode well for the usefulness of this approach to measure and report risks of individual assets and broad portfolios. We believe our approach will also interest corporate decision-makers, both for their own risk analyses and also as an intuitive and relatively easy-to-compute reporting item.

Editor's Note

Submitted 4 June 2022

Accepted 26 September 2022 by William N. Goetzmann

Notes

1. For example, Busch et al. (2022) find that scope 3 estimates are very rarely reported by portfolio companies and third-party estimates are essentially uncorrelated across data providers; Callan (2020) found no respondents who use scope 3 data in investment practice.
2. To identify a company suitable for an illustrative example, we sorted firms on their downstream supply chain exposure and on their own emissions, focusing on those that are at the top of the distribution of the former and at the bottom of the latter. We then analyzed the public statements the companies made to identify the ones that disclosed the importance of customers who may have large climate exposure (e.g., oil majors).
3. See, e.g., Ben-David et al. (2021) and Dai et al. (2022). Offshoring may also be an issue for sovereign emissions.
4. These biases could perhaps be alleviated by leveraging data on revenues from geographic segments and assessing the climate exposure in each region in which a company does business. We believe this approach is coarser than what we propose; in addition, it would require data on climate exposures of specific regions or countries, which itself may pose challenges.
5. For example, a sampling of papers utilizing such data include Cohen and Frazzini (2008), Menzly and Ozbas (2010), Agarwal et al. (2017), or Lee et al. (2019).
6. The jump in early 2010 is driven by adding an additional provider of customer-supplier data; from 2010 onward, we have supply chain linkages data from all underlying providers.
7. While scope 2 emissions show a negative sign in regression (1), this is likely driven by the correlations between scope 1, 2, and 3 emissions. When we regress our measure on scope 2 alone, with or without sector and region controls, we find reliably positive coefficients.
8. We also confirmed a similar finding using MSCI Low Carbon Transition scores (results available by request). We do not report this regression given a shorter time series for that variable (6 quarters of data, as opposed to 36 or 51 quarters for other climate variables we use).
9. (3) and (4) have somewhat different samples because of data coverage, but when (3) is estimated using the same sample as (4), the coefficient on MSCI Climate Change theme is clearly negative, -0.05 (t -statistic = -10.3). Without the region/sector dummies but with all other climate variables from (4), the coefficient is still positive, 0.03 (t -statistic = 11.61).
10. We have repeated all analyses below also with a different source of sentiment data: the Crimson Hexagon series, used in Engle et al. (2020). The results are very similar to what we presented here. We chose to highlight the Ardia et al. data in our paper because the data is publicly available and because it has a somewhat longer time series (the Crimson Hexagon data at our disposal ended in mid-2017).
11. LCT data has a relatively short sample, so we assume a company's historical LCT scores are the same as when MSCI first reports them.

References

- Agarwal, Ashish, Alvin Chung Man Leung, Prabhudev Konana, and Alok Kumar. 2017. "Co-Search Attention and Stock Return Predictability in Supply-Chains." *Information Systems Research* 28 (2): 265–88. doi:10.1287/isre.2016.0656.
- Anderson, Mats, Bolton Patrick, and Frederic Samama. 2016. "Hedging Climate Risk." *Financial Analysts Journal* 72 (3): 13–32. doi:10.2469/faj.v72.n3.4.
- AQR. 2021. "Can Machine Learning Help Manage Climate Risks?" AQR Alternative Thinking Q4: 2021.
- Ardia, David, Keven Bluteau, Kris Boudt, and Koen Inghelbrecht. 2021. *Climate Change Concerns and the Performance of Green versus Brown Stocks*, working paper, National Bank of Belgium.
- Ben-David, Itzhak, Yeejin Jang, Stefanie Kleimeier, and Michael Viehs. 2021. "Exporting Pollution: Where Do Multinational Firms Emit CO₂?" *Economic Policy* 36 (107): 377–437. doi:10.1093/epolic/eiab009.
- Bixby, Steffen, Alfie Brixton, and Lukasz Pomorski. 2022. "Looking Forward with Historical Carbon Data." In *Climate Investing: New Strategies and Implementation Challenges*, edited by E. Jurczenko. London: Wiley and ISTE.
- Bolton, Patrick, Marcin Kacperczyk, and Frederic Samama. 2022. "Net-Zero Carbon Portfolio Alignment." *Financial Analysts Journal* 78 (2): 19–33. doi:10.1080/0015198X.2022.2033105.
- Busch, Timo, Matthew Johnson, and Thomas Pioch. 2022. "Corporate Carbon Performance Data: Quo Vadis?" *Journal of Industrial Ecology* 26 (1): 350–63. doi:10.1111/jiec.13008.
- Callan. 2020. Callan 2020 ESG Survey.
- CFA Institute. 2020. "Climate Change Analysis in the Investment Process."
- Cheema-Fox, Alexander, Bridget Realmuto LaPerla, George Serafeim, David Turkington, and Hui (Stacie) Wang. 2021. "Decarbonizing Everything." *Financial Analysts Journal* 77 (3): 93–108. doi:10.1080/0015198X.2021.1909943.
- Cohen, Lauren, and Andrea Frazzini. 2008. "Economic Links and Predictable Returns." *The Journal of Finance* 63 (4): 1977–2011. doi:10.1111/j.1540-6261.2008.01379.x.
- Dai, Rui, Rui Duan, Hao Liang, and Lilian Ng. 2022. "Outsourcing Climate Change." working paper, ECGI.
- Ducoulombier, Frederic. 2021. "Understanding the Importance of Scope 3 Emissions and the Implications of Data Limitations." *The Journal of Impact and ESG Investing* 2021: jesg.2021.1.018. doi:10.3905/jesg.2021.1.018.
- Engle, Robert, Stefano Giglio, Bryan Kelly, Heebum Lee, and Johannes Stroebe. 2020. "Hedging Climate Change News." *The Review of Financial Studies* 33 (3): 1184–216. doi:10.1093/rfs/hhz072.
- Garvey, Gerald, Mohanaraman Iyer, and Joanna Nash. 2018. "Carbon Footprint and Productivity: Does the 'E' in ESG Capture Efficiency as Well as Environment?" *Journal of Investment Management* 16 (1): 59–69.
- IBM. 2020. "2020 IBM and the Environment Report." https://www.ibm.com/ibm/environment/annual/IBMEnvReport_2020.pdf
- Kolay, Madhuparna, Michael Lemmon, and Elizabeth Tashjian. 2016. "Spreading the Misery? Sources of Bankruptcy Spillover in the Supply Chain." *Journal of Financial and Quantitative Analysis* 51 (6): 1955–90. doi:10.1017/S0022109016000855.
- Lee, Charles, Stephen Sun, Rongfei Wang, and Ran Zhang. 2019. "Technological Links and Predictable Returns." *Journal of Financial Economics* 132 (3): 76–96. doi:10.1016/j.jfineco.2018.11.008.
- Menzly, Lior, and Oguzhan Ozbas. 2010. "Market Segmentation and Cross-Predictability of Returns." *The Journal of Finance* 65 (4): 1555–80. doi:10.1111/j.1540-6261.2010.01578.x.
- Pastor, Lubos, Robert Stambaugh, and Lucian Taylor. 2021. "Dissecting Green Returns." *Working Paper*, University of Chicago.
- Pedersen, Lasse, Shaun Fitzgibbons, and Lukasz Pomorski. 2021. "The ESG-Efficient Frontier." *Journal of Financial Economics* 142 (2): 572–97. doi:10.1016/j.jfineco.2020.11.001.